

STRUCTURAL ENGINEER'S REPORT

Halberd Forensic Engineering · Our reference HFE-2025-0515

INSTRUCTIONS

Instructed by:	Ridgeway Kessler Brokerage Services
On behalf of:	Fairmount Textile Mills Holdings, Inc.
Policy number:	CP-2210-55870
Author:	P. Vanterpool-Aziz, PE, CFEI
Instructed by:	Ridgeway Kessler Brokerage Services
Instructed on:	14 May 2025
Site attendance:	15 May 2025
Scope:	Cause of the collapse and the condition of the remaining structure

BUILDING AND PLANT AFFECTED

Building:	1D — Warehouse and Shipping, 88,000 sq ft, metal frame, 2007
Mezzanine:	1,740 sq ft collapsed, 3,900 sq ft shored and withheld
Racking:	Zone 3, 14 bays destroyed, 6 to inspect
Fire protection:	ESFR, two branch lines and 11 heads sheared
Stock:	Approximately 2,900 rolls, 610 total loss on triage

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Circumstances and findings

CIRCUMSTANCES

Circumstances:

The mezzanine was supplied and installed in 2018 by Foothills Material Handling Systems, Inc. under a design-build purchase order. It carries a posted capacity placard of 125 psf. Below it is the ESFR-protected pick-and-pack area and racking Zone 3.

FINDINGS

Findings:

The failure initiated at a bolted beam-to-column connection at column line C-4. Several bolts were recovered from the debris in a sheared condition and others show elongated holes, indicating the connection was carrying load in bearing on an under-sized bolt group before it let go. Reconciling the warehouse management system's inventory record for the north bay against the collapsed area puts the loading between 180 and 230 psf against the posted 125 psf. Both conditions are present: the connection was inadequate for the design load, and the actual load materially exceeded the design load. The sealed drawings and connection calculations for the 2018 installation have been requested and are not yet produced.

Date of loss:	14 May 2025
Cause of loss:	Collapse — bolted connection failure on a storage mezzanine
Incident reference:	FTM-2025-0514
Injuries reported:	0

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Preliminary quantum and reserve advice

PRELIMINARY QUANTUM

Basis:

The figures below are preliminary and are given for reserving purposes only. They are subject to verification and may move as the investigation continues.

Mezzanine demolition, redesign and reconnection:	USD 1,400,000
Racking replacement and inspection:	USD 380,000
ESFR sprinkler repair and system restoration:	USD 165,000
Building repairs — girts, slab, baseplates, etc:	USD 420,000
Debris removal and shoring:	USD 350,000
Two leased fork trucks:	USD 118,000
Estimated loss:	8,700,000
Currency:	USD

ON THE DEFECT EXCLUSION AND THE ABSENCE OF AN OVERLOAD EXCLUSION

Comment:

The policy excludes loss arising from a defect or error in design or workmanship but writes back resulting physical damage from an insured peril. If the connection design or its installation was defective, the cost of making good the connection is one question and the resulting collapse damage to the racking, the stock and the building is another. Separately, this form carries no express overload exclusion for real property, unlike the contractors equipment forms in the same programme. Both points are recorded for the coverage file.

RECOMMENDATION

Recommendation:

Keep the remaining 3,900 sq ft unloaded and shored; this engineer will not release it. Preserve the sheared bolts, the connection plates and the C-4 column section, and notify Foothills Material Handling Systems before any testing.